Turning Factory Waste into Verified Energy Savings & Potential Carbon Revenue

Industrial Energy Intelligence, Live LLM Copilot (Groq Llama-3.3-70B / Gemini 2.5 Flash), Carbon Credit Forecaster & 3D EPMS Digital Twin for Bangladesh RMG Exporters.

Pilot Facility

Rupsha Garments

Gazipur, Bangladesh • 18 Machines • Seed #20261018

Baseline Power

52,300 kWh

৳ 6,48,750/mo @ ৳12.40/kWh • Scope 1+2: 34.2 tCO_2e/mo

Calibrated EF

0.560

kg CO_2e/kWh (DoE Grid) • 286 MWh = 160.2 tCO_2e

Live AI Copilot

Llama-3.3-70B

+ Gemini 2.5 Flash + Mandatory Human Gate

Why Factories Need Business Profit + Carbon Proof — Not Just Dashboards

1. Hidden Night-Shift & Leak Losses

Sewing lines left idling (4.8–5.4 kW) and compressed air leaks on COMP-01 (+18.5 kW) waste ৳ 68,000+ monthly without automated triage.

2. DPDC Peak Tariff Surges

Peak windows jump to ৳ 14.88/kWh (+20%), inflating electricity bills unless chillers and cutters shift off-peak rapidly.

3. Buyer & EU CBAM Pressure

International apparel brands demand auditable Scope 1 + Scope 2 reduction proof. Owners want clear monthly BDT savings, CapEx payback, and potential carbon monetization.

Real-Time LLM Reasoning + Mandatory Human-in-the-Loop Gate

Live LLM Telemetry Copilot (EN & বাংলা)

- **Groq Llama-3.3-70B-Versatile** primary + **Gemini 2.5 Flash** fallback + deterministic telemetry synthesizer.
- Reads live sensor state during faults (e.g., COMP-01: 31.2 kW / 74.5°C).
- Enforces 3 auditable output blocks: **[REASONING TRACE]**, **[AUTONOMOUS RECOMMENDATION & ROI]**, and **[SAFETY GUARDRAIL]**.

Autonomous Workflow Confirmation Gate

- Pressing **Night Shift Waste**, **Peak Grid Tariffs**, or **Compressor Air Leak** triggers autonomous triage.
- Auto-generates telemetry alert (ALT-AUTO-*), computes CapEx, payback months & 12-month ROI.
- Opens Manager Confirmation Gate before applying any simulated breaker/setpoint change.

6 Financial & Climate Decision Engines on Calibrated Math

1. Low / Med / High Scenarios

Assumed $10, $18, $30/$tCO_2e$ + interactive user price slider ($5–$60) & custom input ($1–$150).

2. 12-Month Savings Trajectory

Chart.js cumulative BDT business return + cumulative avoided tCO_2e trajectory over 12 months.

3. CapEx Payback & 12-Mo ROI

Live CapEx payback months (energy savings only vs. energy + potential carbon credits).

4. Carbon Target Tracker

Shows year-end target attainment probability %, projected annual tCO_2e, and shortfall gap.

5. Buyer Compliance Score

Scores readiness against international brand Scope 3 reduction requirements (LOW / MODERATE / HIGH risk).

6. EU CBAM Tax What-If Slider

Interactive $0–$150/t carbon tax simulator showing avoided liability vs. residual exposure.

Transparency Notice: এটি ধারণামূলক মূল্য, বাস্তব বাজারদর নয় (Illustrative price, not live market price). Bangladesh carbon market requires baseline & MRV verification — reported strictly as **Potential Revenue (সম্ভাব্য আয়)**.

Single Source of Truth: Python, TypeScript, JavaScript & Docker

Full-Stack Repository Structure

- src/agents/copilot_agent.py — Groq/Gemini client + Autonomous Playbook
- src/api/{main,routes}.py — FastAPI REST endpoints + OpenAPI schema
- src/core/calculations.py — Python parity engine
- src/frontend/ & preview/ — Synced SPA + TypeScript domain
- Dockerfile + docker-compose.yml + .github/workflows/ci.yml

Automated Test & Verification Suites

- **Domain Math Suite:** 10/10 verified physics & accounting invariants
- **TypeScript Parity & Typecheck:** 26/26 JS/TS checks + tsc --noEmit
- **E2E Route & Chart Tests:** All 15 SPA routes + Chart.js + Gate workflows
- **FastAPI Pytest Suite:** Live LLM telemetry reasoning & carbon endpoints
- **Headless Chromium Verification:** Zero console errors, 1280px responsive audit

See the Entire Causal Story in Three Clicks

Step 1: Enter Sandbox

Click **"Enter as Manager"** in the 1-Click Demo Access portal on fahad3235.github.io/green-factory-ai.

Step 2: Trigger Scenario

Press **Peak Grid Tariffs** or **Night Shift Waste** on the Dashboard Business Impact Card. Watch alert, ROI & Confirmation Gate trigger live.

Step 3: Inspect AI & Carbon

Approve Gate, switch to **CarbonIQ Report** for 12-mo profit chart + CBAM slider, or ask Copilot about **COMP-01 Air Leak**.